20th International Conference on Advances in ICT for Emerging Regions (ICTer 2020): 294 - 295

Assistive Learning Platform for Children with Down Syndrome

Sahan Wellala[1], S.A.Thathsarani[2], Dishan Senaratne[3], Pavan Samaranayake[4], Anuradha Jayakody[5]
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
[1]sahanwellala@yahoo.com , [2]thathsarani001@gmail.com, [3]dishan507@gmail.com , [4]pavansamaranayake@gmail.com , [5]anuradha.j@sliit.lk

2020 20th International Conference on Advances in ICT for Emerging Regions (ICTer) | 978-1-7281-8655-9/20/$31.00 ©2020 IEEE | DOI: 10.1109/ICTer51097.2020.9325487

***Abstract*— Down Syndrome (DS) is a genetic disorder. Researchers believe that persons with DS generally have poor logical knowledge, communication, motor skills, and skills needed for everyday life. The proposed system is a web-based assistive learning platform for children with DS to address those problems. It provides an excellent opportunity for learning educational subjects, including math and language. The authors also created interactive modules improving their health habits, social skills, and motor skills. Since the authors deeply consider their requirements, the system was developed by providing excellent features with those modules. Users' faces can be recognized to keep their attention with the system, thereby suggesting and referring their most interesting content according to their emotions while using the system. Also, the dashboard can analyze user data. Most importantly, it capable of assessing the users through the system. Here the research team has assessed 50 children with DS, and 31 showed improvement after using the system. Therefore, the proposed system with all these modules and features can be introduced as a very productive assistive learning platform in Sri Lanka.**

Keywords*— Down syndrome, assistive technology, motor skills, scaling tools

I. Introduction

Down Syndrome (DS) [1] is a genetic disorder that causes delays in developing the body and the brain[2]. The most common physical characteristics are low muscle tone, slanted eyes, and round face. These characteristics are not aspects of the disease; however, lifetime conditions underlined for having difficulties such as auditory, visual, and communication are caused to have poor learning rates of children with Down Syndrome (DS) [2].

The education of children with down syndrome is the least studied area in the world [3], and with the lack of studies, the research team acknowledged that there is no proper system exists to assess the children with DS in Sri Lanka. Although the children's teaching process with DS is different; therefore, the teachers and parents must be strongly oriented towards them. Nevertheless, most of these children tend to be refused by society and burden for their families due to the apathetic and informal way of teaching these children.

However, studies had identified the effectiveness of computer-assisted teaching over the traditional teaching of children with DS [4]. The satisfying results acquired from the earlier studies are clear evidence for a potential further study that will benefit the learning system for the children with DS. Although accompanied by modern technologies will be sufficient for the proposed solution, thus refine the quality of the product. Currently, there is no system developed considering these details.

Influenced by the studies, the proposed system built to improve and assess children's learning skills with DS. It also embedded with cutting edge technologies to enhance the user experience of the system. Furthermore, the system provides its content in the native language, "Sinhala" by identifying the users' basic needs in Sri Lanka.

II. System Overview

The web-based application consists of learning modules targeting four different areas that the authors identified when doing the background studies. The learning content comprises improving the language, mathematics, social, motor learning skills, and module to teach healthy habits. These learning modules contain interactive activities, quizzes, and multimedia items related to each skill. Children's performances and the progress in the activities can be viewed using the implemented doctor dashboard module. It analyzes how many activities each child attempted in each learning module, what learning activities were difficult for them to do, and what scored low. Doctors, guardians, or teachers of the children can see these analyzed details.

There are exclusive implemented features to improve the users' engagement with the system. A Facial recognition module is developed to detect the presence of the user. When the child is not present, it will notify by alert using an animated character. Moreover, the module also identifies the user's facial emotion and helps to understand the user's current mood. A recommendation system is built to suggest the content according to the user data. Simple alert displays and suggest the user switch the content which contained educational and entertainment activities. It is triggered when the user's emotional status is getting negative, which is calculated by the facial recognition module's emotional data. The authors initially develop the system's underlying features to enhance the system's learning process for children with down syndrome.

III. Methodology

All the learning is done through web-activities in learning modules created according to the requirements of children with Down Syndrome at three centers. It made it possible to develop these in a way that would be more suitable for children. These activities are developed based on React and use Node and Mongo to manage the backend.

Other than the learning modules, emotion recognition through the system has two main steps: detecting the face and capturing the emotions. Face detection performs two main functions. One is capturing the face to see if the user is in front of the screen—the face detection module functions

978-1-7281-8655-9/20/$31.00 ©2020 IEEE

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:53:52 UTC from IEEE Xplore. Restrictions apply.

using the face-api.js JavaScript library to detect the faces [5]. A photo was taken every 3 seconds and fed into the face detection module. Accordingly, the face-detection module captures the faces and identify the face landmarks. Face landmark refers to identifying facial features. Next, it identifies the coordinates of the face location in the input image frame. After that, it extracted only the face from the input image frame with the help of the previously identified face coordinates and passed it into the emotion recognition module to detect the emotion out of it. The emotion detection module starts its execution with the extracted face image from the previously discussed face detection module. Then the image is resized into the size of 96x96 pixels. Next, the image is converted to grayscale and applies max normalization. Next, the normalized image is fed into the Deep Convolutional Neural Network model. In the model, "ELU" has been used as the activation function because it performed well when compared with the "LeakyRelu" activation function, and "he_normal" has been used as the kernel initializer. A black and white image data set which contains images over 21000 were used to train the model. Those images were 48x48 pixels in size and labeled according to the emotion as happy, sad, and neutral. Then the result is sent back to the application for the further usage of the content switching module. When creating the model, "TensorFlow 2" and "Keras" were used. Moreover, the model was developed in the "Python 3.8" environment.

Moreover, the content switching module of the system triggers when the user's emotional status is negative. A function is implemented to calculate the specific period's emotional status considering the negative emotions (sad, fearful, angry) getting from the user. A machine learning recommendation algorithm is implemented to generate the suggestive content for each user considering their preferences and content metadata. Due to the difficulty of getting feedbacks for the content data from the children with DS, the count of each user-content interaction data is scaled and used as ratings. The algorithm is built as a model of Hybrid Matrix Factorization [6]. It is implemented using the "LightFM" python library [7], which uses a very efficient implementation of combined ML algorithms.

IV. Results and Discussion

The proposed system is tested with 50 children with Down Syndrome children studying in special education institutes. They are namely "Senehasa" Children Resource Center, "Eksath" Children Resource Center, and the "Methnana" Foundation in Sri Lanka and 31 of the children showed improvement in skills according to the CS-DS scale [8]. When comparing the students' scores before using the platform and after using the platform for two months, out of the fifty students, thirty-one students improved. Therefore the proposed system has assisted children with Down Syndrome to improve their learning skills and cognitive abilities.

When considering the core of the proposed assistive learning platform, it integrates with another set of modules to improve its interactivity. The face detection module that detects the child's presence performed well with the ambient light, and the authors identified that it was difficult to detect the faces when the sunlight directly hits the web camera. The next module is the emotion detection module used to identify the child's emotional changes while engaged in the activities. The DCNN model used to identify the emotion achieved an overall 83% accuracy on the test set. The dataset that is used when creating the model consisted of labeled images of over thirty-five thousand. The model predicted emotion under three classes, happy, sad, and neutral, and achieved precisions of 93%, 81%, and 71% in those classes, respectively. The availability of the images for the separate classes resulted in different precisions in the classes.

Finally, based on the emotions and the user activity within the application, the application suggests activities. These include favorite quiz activities, music, and entertaining videos. The recommendation model achieved 89% accuracy with the WARP as the loss function

V. Conclusion

When considering the results, the proposed system can be introduced as a very productive assistive learning platform in Sri Lanka. In the future, the research team is planning to extend the learning areas further and plans to optimize the application for web and mobile platforms. Moreover, the team plans to introduce applications for special education centers throughout the country to provide more learning opportunities for children with down syndrome.

References

[1] C. J. Epstein, "Down Syndrome," in *Abnormal States of Brain and Mind*, Boston, MA: Birkhäuser Boston, 1989, pp. 43–44.

[2] S. Ivic, "Communication Skills of a Child with Down Syndrome at the End of the First Grade of Elementary School," *J. Educ. Pract.*, vol. 7, no. 23, pp. 19–28, 2016, [Online]. Available: http://web.a.ebscohost.com/ehost/detail/detail?vid=6&sid=cd17a7ef-1807-4f3a-9c4b-4ef5a9fd12f9%40sessionmgr4007&hid=4112&bdata=JnNpdGU9ZWhvc3QtbGl2ZQ%3D%3D#AN=EJ1112911&db=eric.

[3] J. M. Ortega-Tudela and C. J. Gómez-Ariza, "Computer-assisted teaching and mathematical learning in Down Syndrome children," *J. Comput. Assist. Learn.*, vol. 22, no. 4, pp. 298–307, 2006, doi: 10.1111/j.1365-2729.2006.00179.x.

[4] J. Feng, J. Lazar, L. Kumin, and A. Ozok, "Computer usage by children with down syndrome: Challenges and future research," *ACM Trans. Access. Comput.*, vol. 2, no. 3, pp. 1–44, Mar. 2010, doi: 10.1145/1714458.1714460.

[5] GitHub, "face-api," 2020. https://github.com/justadudewhohacks/face-api.js (accessed Jan. 02, 2020).

[6] M. Kula, "Metadata embeddings for user and item cold-start recommendations," *CEUR Workshop Proc.*, vol. 1448, pp. 14–21, 2015.

[7] "LightFM — LightFM 1.15 documentation." https://making.lyst.com/lightfm/docs/lightfm.html (accessed Sep. 13, 2020).

[8] C. M. Startin, E. Rodger, L. Fodor-Wynne, S. Hamburg, and A. Strydom, "Developing an informant questionnaire for cognitive abilities in down syndrome: The Cognitive Scale for Down Syndrome (CS-DS)," *PLoS One*, vol. 11, no. 5, May 2016, doi: 10.1371/journal.pone.0154596.

Authorized licensed use limited to: University of New South Wales. Downloaded on December 02,2025 at 14:53:52 UTC from IEEE Xplore. Restrictions apply.